

Django Project 1 — Library Management System

Objective

Build a web-based library management system where students can browse and borrow books, while librarians manage books and borrowing records.

User Roles

- Students can register/login, browse books, search for books, view details, borrow available books, view current loans and borrowing history, and return books.
- Librarians can add, edit and delete books, view students, monitor borrowing records, and manage returns.

Suggested Models

- Book: title, author, ISBN, category, published date, quantity, available quantity.
- BorrowRecord: student, book, borrowed date, due date, returned date, and return status.

Business Rules

- A student cannot borrow an unavailable book.
- A student cannot borrow the same book while they already have it.
- A student can have a maximum of 3 active borrowed books.
- Borrowing decreases available quantity; returning increases it.
- Only authenticated users can borrow or return books.
- Librarian operations must be protected from normal students.

Suggested Pages

Login/Register, Student Dashboard, Book List, Book Details, Search, My Borrowed Books, Borrowing History, Librarian Dashboard, Book Management, Student List, Borrow Records

Bonus Features

- Due dates and overdue status
- Fine calculation
- Pagination
- Category filtering
- Book cover uploads
- Book ratings/reviews

Technical Requirements

- Use Django and the Django ORM.
- Use Django Templates with HTML/CSS.
- Use Django Authentication and appropriate access control.
- Use SQLite during development.

- Use Django Forms or ModelForms where appropriate.
- Do not use React, Django REST Framework, or another frontend framework for the main project.

Assessment

Area	Marks
Models & relationships	15
Authentication & authorization	15
CRUD functionality	15
Django ORM	15
Business logic	20
Templates/UI	10
Search/filtering	5
Code organization	5
Total	100